Experiment 3: Verification of Ohm's Law

Aim: To verify Ohm's law and to determine the resistance of a given wire.

Apparatus: Battery eliminator, ammeter, voltmeter, rheostat, resistance wire, one way key, connecting wires and sandpaper.

Theory: Ohm's law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided the physical conditions such as temperature remain constant. Therefore V is proportional to I, and the constant of proportionality is the resistance R of the conductor.

Procedure: Clean the ends of the connecting wires with sandpaper. Connect the circuit as shown in the diagram, keeping the ammeter in series and the voltmeter in parallel with the resistance wire. Insert the key and adjust the rheostat so that a small current flows. Note the readings of the ammeter and the voltmeter. Repeat the observation for five different values of the current by adjusting the rheostat. Record all readings in the observation table.

Observations: The ratio of the potential difference to the current was found to be very nearly constant for every reading taken.

Result: Ohm's law is verified. The resistance of the given wire was found to be constant within the limits of experimental error.

Precautions: All connections should be tight and clean. The key should be inserted only while taking readings so that the wire does not heat up. The ammeter and voltmeter should be checked for zero error before beginning.